

UNIVERSITY OF EDINBURGH
COLLEGE OF SCIENCE AND ENGINEERING
SCHOOL OF INFORMATICS

COMPUTER COMMUNICATIONS AND NETWORKS

Thursday 20 August 2009

14:30 to 16:30

Year 3 Courses
Convener: K Kalorkoti
External Examiners: A Frisch, J Gurd

INSTRUCTIONS TO CANDIDATES

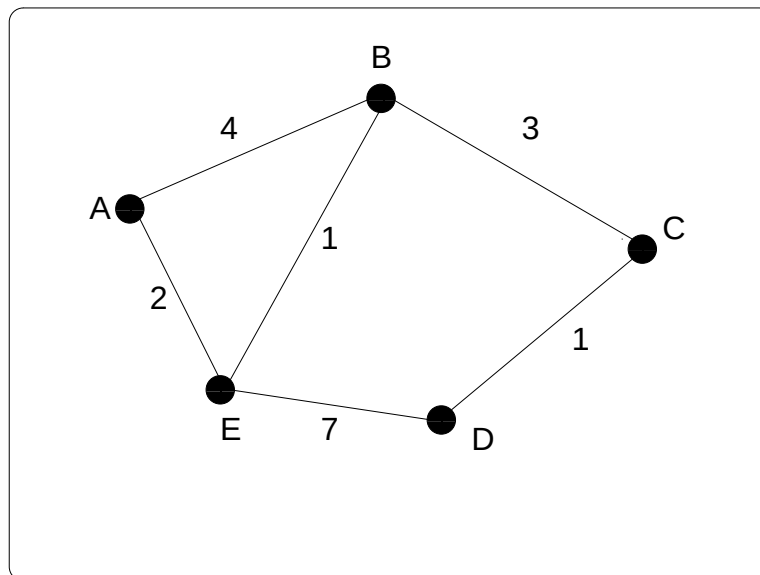
Answer any TWO questions.

All questions carry equal weight.

CALCULATORS MAY BE USED IN THIS EXAMINATION

1. (a) Define “protocol.” [2 marks]
- (b) Give two advantages of protocol layering. [2 marks]
- (c) A system has an n -layer protocol hierarchy. Applications generate messages of length M bytes. At each of the layers, an h -byte header is added. What fraction of the network bandwidth is filled with headers? [2 marks]
- (d) Give two key differences between the Internet (TCP/IP) and OSI network architectures. [2 marks]
- (e) Give two main benefits of web caching. Give an example of another application layer protocol that employs a similar caching idea. [3 marks]
- (f) Consider distributing a file of size $F = 10$ Gb to N peers. The server has an upload rate of $u_s = 50$ Mbps, and each peer has a download rate of $d_i = 2$ Mbps and an upload rate of u . For $N = 10$ and 100 , and $u = 250$ Kbps and 500 Kbps, tabulate the minimum distribution times for each of the combination of N and u for both client-server distribution and P2P distribution. [8 marks]
- (g) Consider sending a large file from one host to another over a TCP connection that has no loss. Suppose TCP uses AIMD for its congestion control *without* slow start. Assuming $CongWin$ increases by 1 MSS (maximum segment size) every time a batch of ACKs is received and assuming approximately constant RTTs (round-trip times), how long does it take for $CongWin$ to increase from 1 MSS to 7 MSS (assuming no loss events)? What is the average throughput (in terms of MSS and RTT) for this connection over that period from the previous question? [4 marks]
- (h) Describe two scenarios where use of standard TCP can result in unfairness. [2 marks]

2. (a) List two advantages and two disadvantages of having international standards for network protocols. Name two prominent sets of standards relevant to computer networking. [6 marks]
- (b) An object of size S needs to be transferred over a 2-Mbps long-distance link with a round-trip propagation delay of 200 ms. At what value of the transfer size S will the transmission delay start to dominate in the overall transfer delay. Assume delays other than due to propagation and transmission to be negligible. [3 marks]
- (c) Suppose a host has a 1-MB file that is to be sent to another host over a 2-Mbps link with 10 ms one-way propagation delay. The file takes 1 second of CPU time to compress 30%, or 2 seconds to compress 50%. Ignore queueing delays.
- Calculate the bandwidth at which the combined time taken for compression and transmission for each of the compression options equals that of the uncompressed case. [4 marks]
 - Does propagation delay also get affected? Why or why not? [2 marks]
- (d) The figure shown below indicates the state of the network topology and link costs at node A at the end of flooding-based link state dissemination (i.e., using link-state broadcast algorithm). Determine the shortest path from node A to every other node using Dijkstra's shortest path algorithm. Show all steps of the algorithm.



[4 marks]

QUESTION CONTINUES ON NEXT PAGE

QUESTION CONTINUED FROM PREVIOUS PAGE

- (e) Give a simple heuristic for finding two paths through a network from a given source to a given destination that can survive the loss of any communication line (assuming two such paths exist and routers never crash). [4 marks]
- (f) How is the notion of “subnets” related to hierarchical routing? [2 marks]

3. (a) Compare channel partitioning and random access MAC (medium access control) protocols at high and low loads. Which of these two classes of protocols permit statistical multiplexing and why? [7 marks]
- (b) List four differences between routers and switches. [4 marks]
- (c) Give two reasons why networks might use an error-correcting code instead of error detection and retransmission. [4 marks]
- (d) An upper-layer packet is split into 5 frames, each of which has an 80 percent chance of arriving undamaged. If no error control is done by the data link protocol, how many times must the message be sent on average to get the entire message through? [4 marks]
- (e) Two CSMA/CD stations are each trying to transmit long (multiframe) files. After each frame is sent, they contend for the channel, using the binary exponential backoff algorithm. What is the probability that the contention ends on round k , and what is the mean number of rounds per contention period? [6 marks]